DevFest

Chennai

Local AI:
The Google Gemma Edition

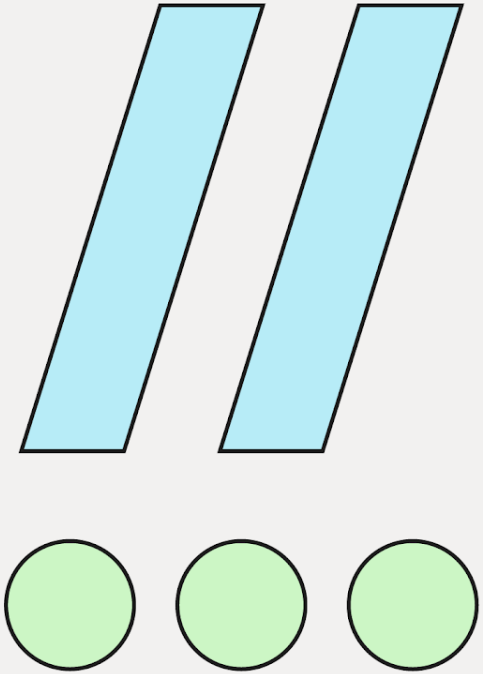
Gokulbarath SK (gokulbarath.is-a.dev)
Founder. BGO APPS . (www.bgoapps.xyz)

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What if your AI never needed the internet?

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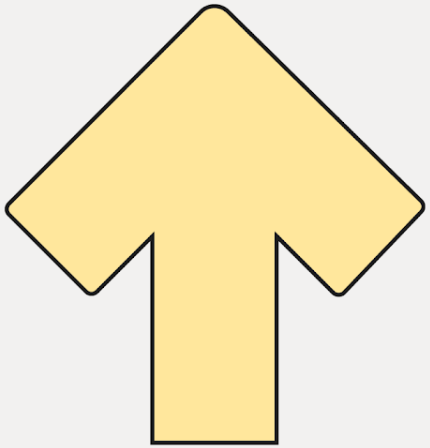
DevFest 2026



01

Local AI

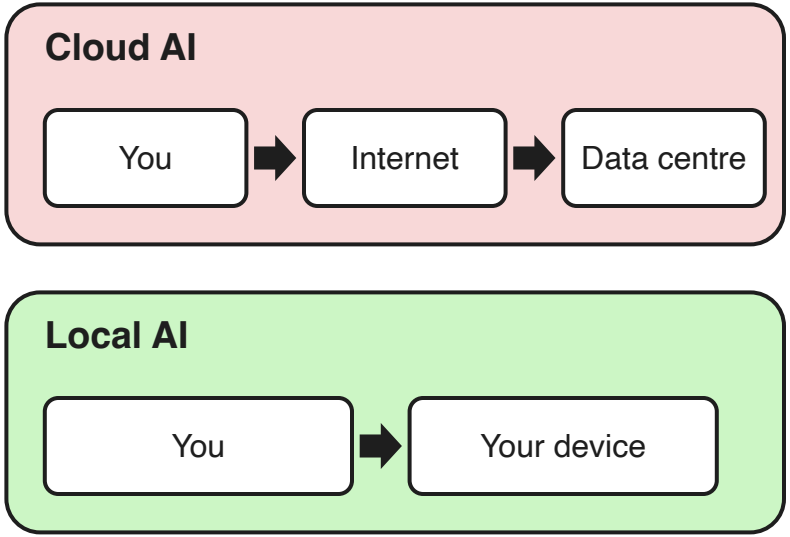
What it is, and why it matters now



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What is local AI?

- The model's weights live on your own hardware
- Prompts and answers never leave the device
- Works with Wi-Fi switched off



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Why run it locally?

Privacy
Your data never leaves the device

Offline
No signal? It still answers

Cost
No per-token bill, ever

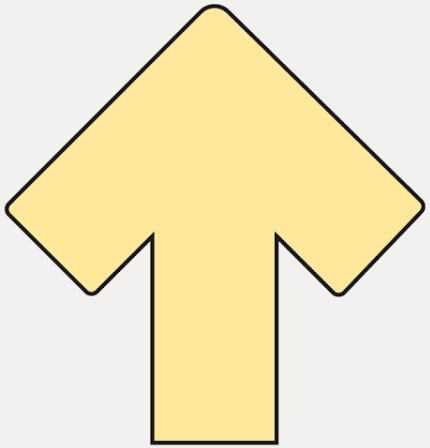
Control
You choose the model, version and rules



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02

Meet Gemma 4



Google's open model family, Apache 2.0

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How an LLM answers: one token at a time

Text → tokens → Attention layers → Score every next token → Pick one, repeat

Input: “The capital of France is”

P(next token | everything so far)

Same spirit as Bayes: update a guess with new evidence.

No lookup, no database: just very good guessing, repeated.



Illustrative probabilities

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Gemma 4 is truly open

- Apache 2.0: free as in total freedom
- No research-only clause
- No “don't make money or we'll sue” clause
- Fine-tune it, ship it, sell it



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The Gemma 4 family

Model	Parameters	Q4_0 memory	Sweet spot
E2B	2.3B effective	2.9 GB	Phones, Raspberry Pi
E4B	4.5B effective	4.5 GB	Pi 5, laptops
26B A4B	26B total / 3.8B active (MoE)	14.4 GB	One consumer GPU
31B	31B dense	17.5 GB	24 GB GPU

Green = what fits a Pi. All Apache 2.0 and multimodal. A 12B sits in between. “E” = effective parameters.



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Decoding the names: E2B, A4B, it, QAT

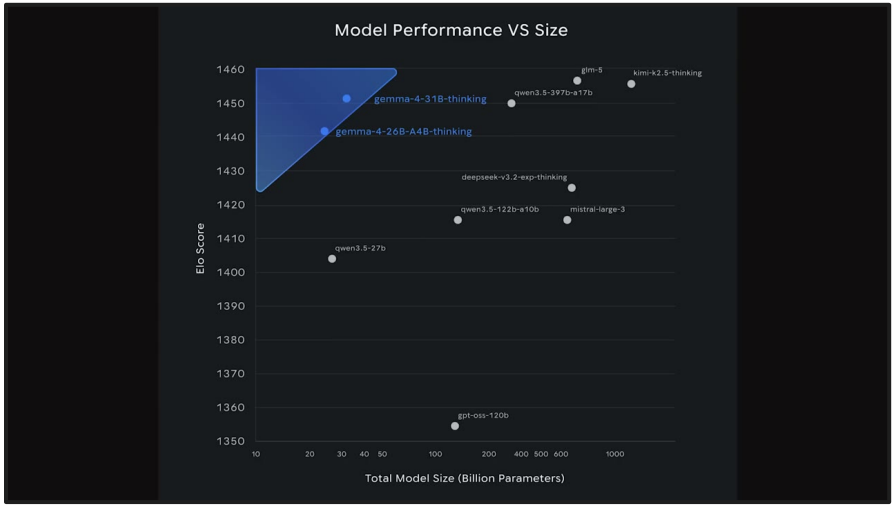
Tag	What it means	Example
E2B · E4B	Effective parameters: the per-layer tables are just lookups	E2B: 2.3B effective, 5.1B stored
26B A4B	Mixture of Experts: 26B total, ~4B Active per token	A router picks a few experts
31B · 12B	Dense: every parameter works on every token	Biggest and most capable
it	Instruction-tuned, chat-ready (base = raw pretrained)	gemma4:e4b-it-qat
QAT	Quantization-aware training: built to survive 4-bit	Google ships QAT checkpoints
Q4_0	4-bit weight format, about 4.5 bits per weight	E2B: 11.4 GB to 2.9 GB

B = billions of parameters. Names follow Google's Gemma 4 docs.



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Small, but not weak



- Top-left is the sweet spot: high score, few parameters
- Gemma 4 31B and 26B A4B sit up there
- E2B and E4B go further: phone and Pi class

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Gemma vs the open-model field

Family	Licence	Local story
Llama	Community licence, extra usage terms	Strong, large
GPT-OSS	Apache 2.0	Bigger models
Qwen · Kimi · DeepSeek	Open weights, terms vary	Frontier-scale, heavy
Gemma 4	Apache 2.0, no strings	E2B / E4B run on a Pi

20 GB
Gemma 4 31B download

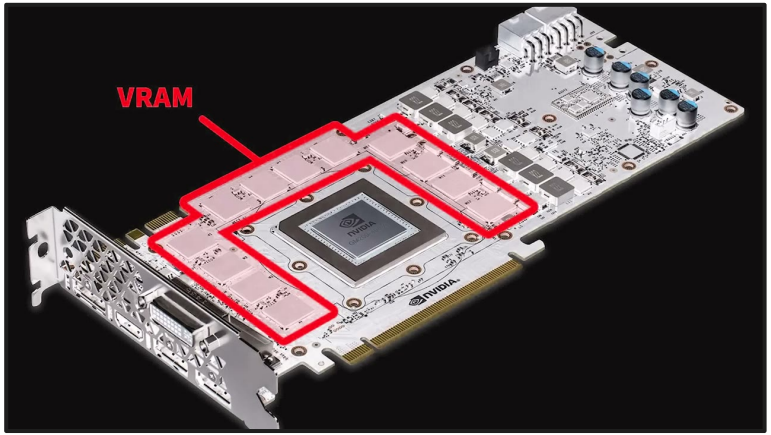
600+ GB
Kimi K2.5, a rival of similar benchmark class



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The real bottleneck: memory, not compute

- Every token re-reads the model's weights from memory
- Speed is about memory bandwidth ÷ bytes read
- Parameter count alone is the wrong yardstick



Pi 5 back-of-envelope: 17 GB/s ÷ ~1.3 GB per token ≈ 13 tok/s ceiling. We measure ~8.

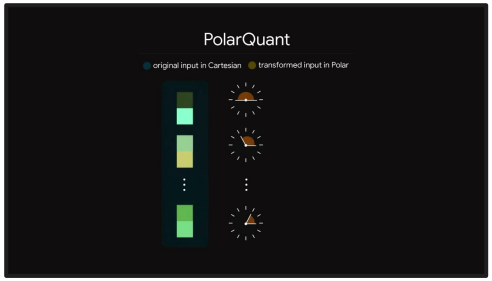
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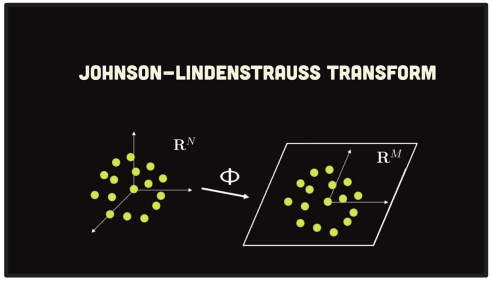
TurboQuant: shrinking the KV cache

1. **PolarQuant**
   Cartesian to polar: store radius and angle, skip normalisation overhead

2. **Johnson–Lindenstrauss**
   Squash to sign bits (+1 / −1) while keeping distances

3. **~3 bits per value**
   Google Research, March 2026. It targets the KV cache, not the weights

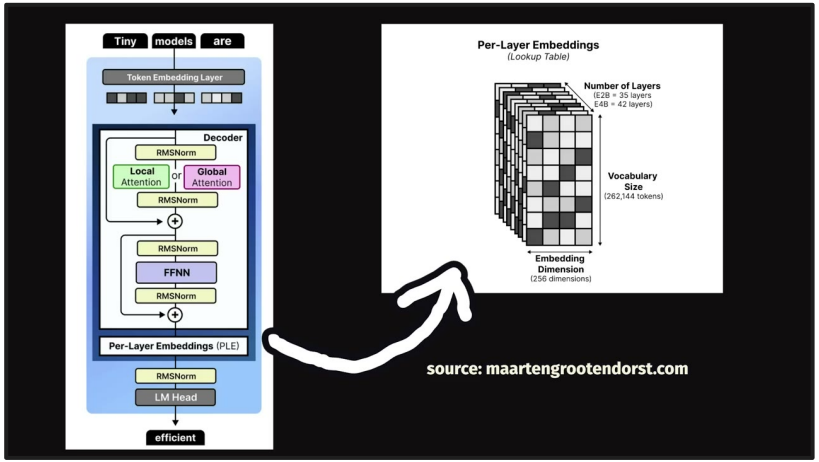




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Per-layer embeddings: the “E” in E2B

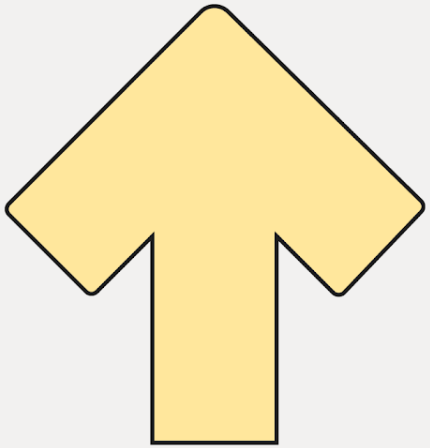
- **Normal:** one embedding at the input, carried through every layer
- **PLE:** each layer gets its own small lookup for every token
- **E2B table:** 262,144 tokens × 35 layers × 256 dims, just lookups, fine in flash
- **So:** 2.3B effective, 5.1B stored



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03

Gemma on a Raspberry Pi 5



Always on, a few watts, fully offline

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Pick the model the Pi can hold

Raspberry Pi 5 · 8 GB

- 4× Cortex-A76 @ 2.4 GHz
- 8 GB LPDDR4X (~17 GB/s)
- CPU inference, no GPU

E4B on 8 GB works but RAM maxes out (7.3 of 7.9 GB, our tests). E2B leaves headroom.



Pi 5: 8 GB RAM, shared with the OS

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Gemma on the Pi in 3 commands

```
$ curl -fsSL https://ollama.com/install.sh | sh
$ ollama pull gemma4:e4b-it-qat
$ ollama run gemma4:e4b-it-qat
```

Ollama handles download, quantised weights and a local API on port 11434.

Gotchas we hit

- It's a thinking model: call /api/chat with think: false or you get empty replies
- Cold load takes ~100 s: warm it up before you go on stage
- Set keep_alive so it stays loaded



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04

Gemma + Robotics

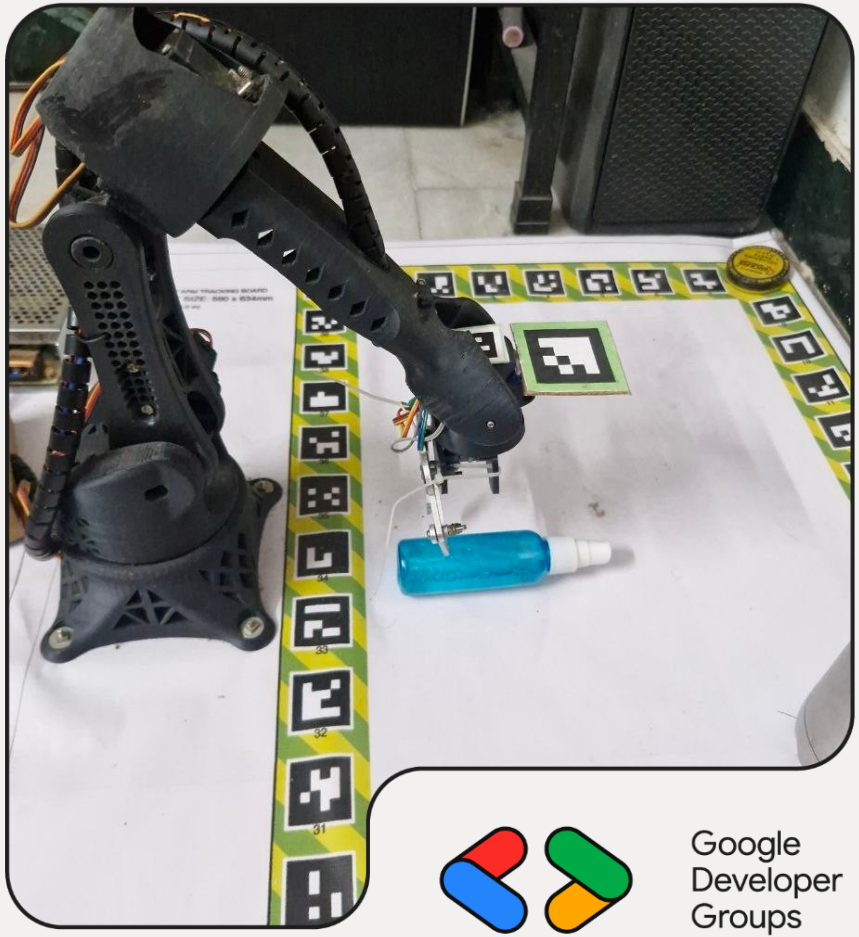


Say it. Gemma sees it. The arm does it

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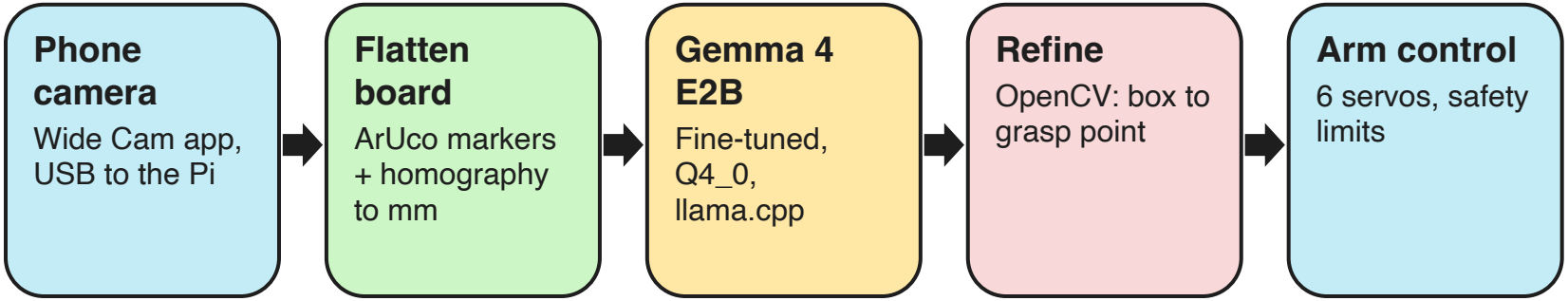
An offline robot arm

- Phone camera sees the board
- Gemma 4 finds the object
- 6-servo arm picks and places
- All on a Raspberry Pi 5, no cloud



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How it works: five stages, one Pi

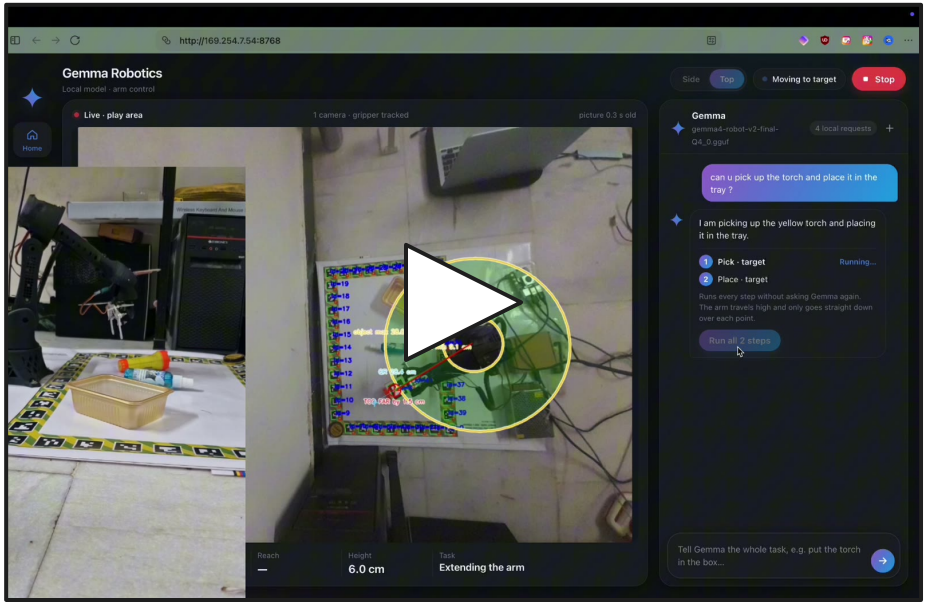


You: “pick up the torch and place it in the box”

Gemma: [{"do":"pick","box":[ymin,xmin,ymax,xmax]}, …] + one friendly sentence

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Demo: Pi-hosted Gemma controlling the arm

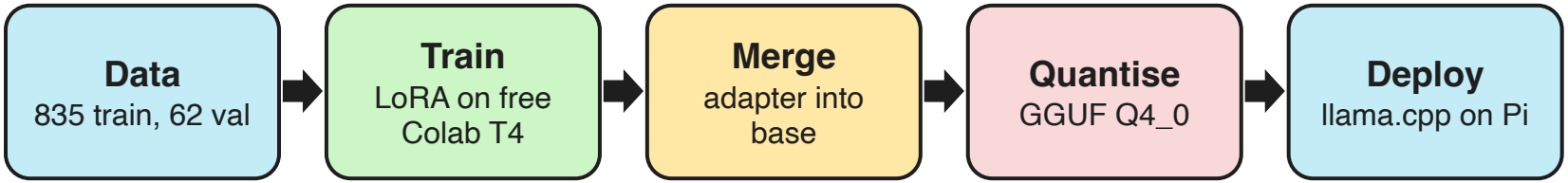


What to watch

- Request becomes boxes
- Base turns, reaches, grips, lifts, places
- No Wi-Fi needed

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Teaching Gemma to point: LoRA fine-tuning



Why LoRA?

Freeze the base model, train tiny low-rank adapters (rank 16) on attention and MLP layers. A checkpoint is ~50 MB.

What we kept fixed

Vision encoder stays as Google made it. Loss only on the answer tokens, at the same 140 image tokens the Pi uses.

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What one training example looks like

image top-down board photo

user “Pick up the black flashlight.”

model [{"do":"pick",
"box":[412,380,531,455]}]
“Got it, grabbing the torch!”

boxes are 0 to 1000 on the image (values illustrative)

835
train examples

62
val, 4 unseen scenes

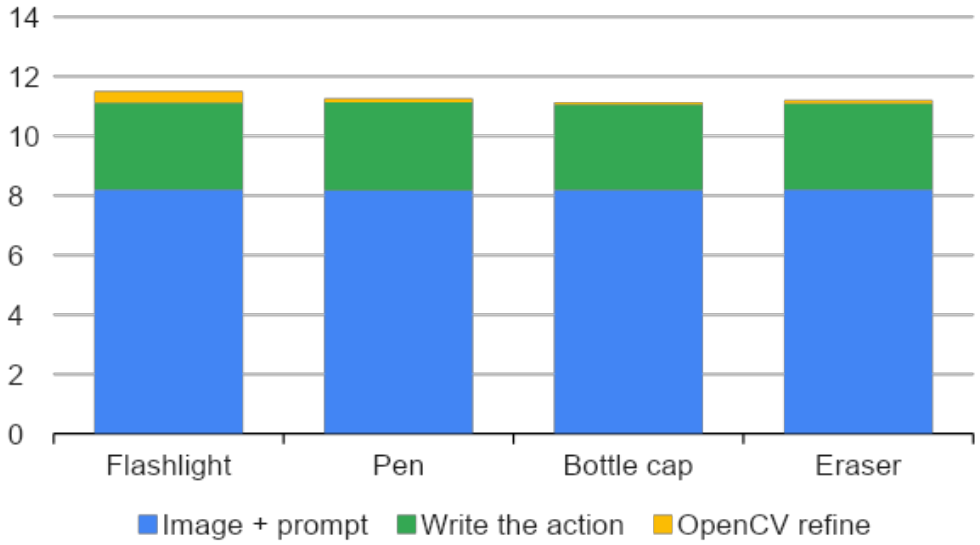
4×
views: flip, 180°, zoom + light

self
replies written by Gemma

Boards photographed from a real phone-camera frame; labels reviewed, 5 fixed by hand

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Benchmark: how fast on a Pi 5?



11.3 s

request to CV-checked grasp point

8 s

is just the image + prompt (140 tokens)

1.3 s

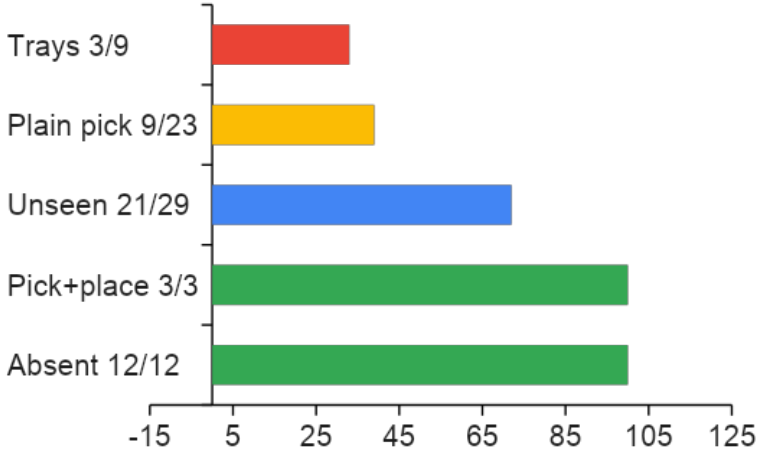
same model on an M4 Mac (median)

Pi 5, E2B Q4_0, 4 threads, no throttling; 4 single-object requests. Model load 4 s excluded

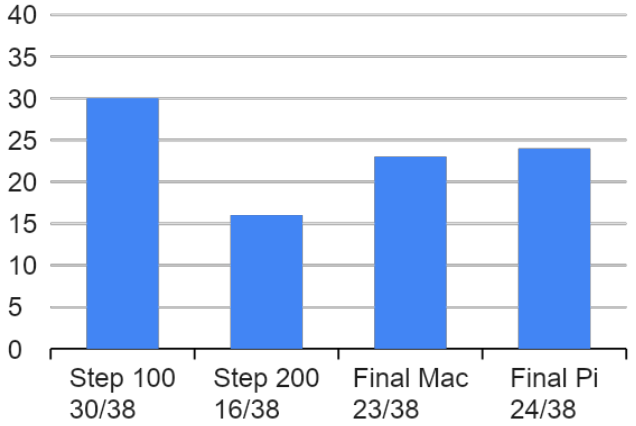
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Benchmark: how often is it right?

On the Pi: 24 of 38 exam prompts pass (63%)



Accuracy swung between checkpoints



38-prompt dev exam; slices overlap. Development numbers, not a published benchmark

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Gemma lessons

Thinking model: set think: false or replies come back empty.

Image tokens set latency: 140 is ~12 s, 280 is ~31 s.

Test every checkpoint: 30, then 16, then 23 of 38.

Teach it to say no: absent object means empty action.



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05

The Future



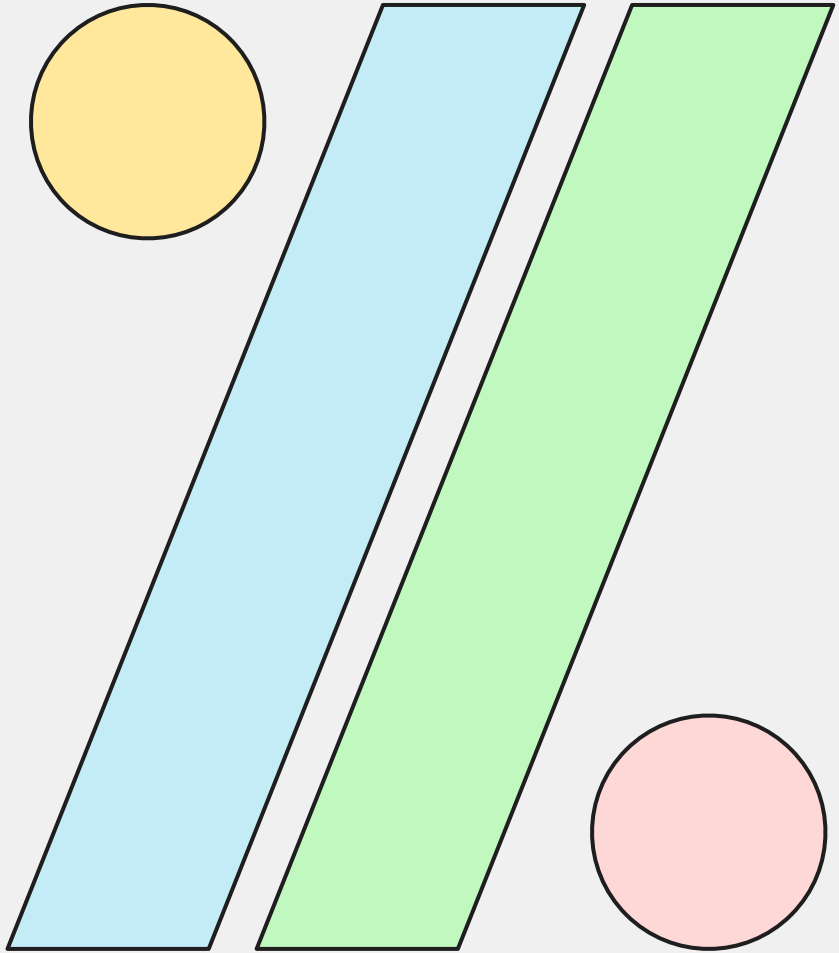
Local brains for robots that can't phone home

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Earth to Moon, one way

1.3 s

Earth to Mars: 3 to 22 min



From a desk to a multi-planetary builder

Today
An arm on a desk, Gemma on a Pi

→

Next
Home, farm and disaster-zone robots that think on-device

→

The Moon
Robot swarms building habitats and pads from lunar soil

→

Mars and beyond
Autonomous crews prepare bases before humans land

What has to be true: small offline models, tiny power budgets, self-calibration, swarm coordination, safety first.

A vision, not a forecast. NASA and ESA already study 3D-printing habitats from lunar regolith



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Thank you!

Questions?

Slides, demo video & PDF:

gokulbarath.is-a.dev/devfest2026